Supporting Material for:

Three-Dimensional Contact Topology Predicts Kirchhoff Contact Laplacian Fundamental Frequencies Across 1,040 Protein Structures

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SUPPORTING TEXT S1: RAYLEIGH QUOTIENT & RIGIDITY PERCOLATION SCALING DERIVATION

By the Courant-Fischer-Weyl variational principle, the smallest non-zero Fiedler eigenvalue λ_1 of the $N_{C\alpha} \times N_{C\alpha}$ Distance-Weighted Kirchhoff Contact Laplacian Γ satisfies $\lambda_1 = \min_{\mathbf{u} \perp \mathbf{1}} (\mathbf{u}^T \Gamma \mathbf{u})/(\mathbf{u}^T \mathbf{u})$. Selecting a trial displacement aligned with the principal radius of gyration coordinates $(\mathbf{r}_i - \mathbf{r}_{\text{cm}})$ yields the Rayleigh upper bound $\lambda_1 \leq (\langle z \rangle R_c^2)/(2R_g^2)$. In folded globular proteins, interior non-bonded coordination $\langle z \rangle$ exceeds the Maxwell rigidity percolation threshold ($\langle z \rangle_c \approx 6.0$), stiffening the effective shear modulus $E(\langle z \rangle) \sim \langle z \rangle^{2b}$ and producing the dimensionless empirical power law $\nu_1(\text{THz}) = a_0(\langle z \rangle/(R_g/R_0))^b$ with $a_0 = 0.42338$ THz, $R_0 \equiv 1.0$ Å, and $b = 1.1194$ across $N = 1,040$ structures.

SUPPORTING TABLE S1: REPRESENTATIVE 11-PROTEIN CANONICAL BENCHMARK COHORT

PDB ID	Biomolecule & Gene	UniProt	$N_{C\alpha}$	R_g (Å)	$\langle z \rangle$	$\langle z \rangle/(R_g/R_0)$	dw-GNM ν_1
2FY2	Choline O-acetyltransferase (CHAT)	P28329	607	25.07	10.50	0.4189	0.0725 THz
5O3L	Tau Paired Helical Filament (MAPT)	P10636	73	24.84	5.92	0.2382	0.0643 THz
1IYT	Amyloid-beta Peptide 42 (APP)	P05067	42	16.01	6.57	0.4104	0.1190 THz
1XQ8	Alpha-Synuclein Micelle-Bound (SNCA)	P37840	140	38.68	6.14	0.1588	0.0457 THz
2F8A	Glutathione Peroxidase 1 (GPX1)	P07203	184	15.51	10.17	0.6560	0.2449 THz
1PU0	Cu/Zn Superoxide Dismutase (SOD1)	P00441	153	14.57	9.82	0.6736	0.2512 THz
1UCU	Salmonella Flagellin (fliC)	P06179	405	44.78	8.59	0.1918	0.0372 THz
5MHT	Candida Adhesin Domain (ALS)	Q59XX2	300	19.68	10.45	0.5308	0.1219 THz
1E9Z	Helicobacter pylori Urease Beta (ureB)	P69996	569	24.84	10.55	0.4247	0.0882 THz
6R48	Protochlorophyllide Reductase (PORA)	Q42536	298	18.37	10.50	0.5718	0.1651 THz
2XSN	Tyrosine 3-Monooxygenase (TH)	P07101	335	19.99	9.69	0.4848	0.1191 THz

SUPPORTING TABLE S2: PUBLIC DATA & CODE AVAILABILITY MANIFEST (SEED = 20261018)

Public Repository Artifact Path	Size / Scope	Description / Verification Scope
pdb_1000_cohort_dataset.csv	211,163 B (N = 1,040)	Full 1,040-structure geometric & dw-GNM table
pdb_1000_cohort_validation.json	816,011 B (N = 1,040)	Complete regression, CV, OOS & null benchmarks
pdb_115_cohort_dataset.csv	27,643 B (N = 115)	Primary 115-protein validation table
fig1_n1040_headline_scaling_law.png	600 DPI White-BG	4-Panel Publication Figure 1 (PNG / PDF / TIFF)
scripts/kirchhoff_laplacian_solver.py	Python Module	Exact dw-GNM Kirchhoff Laplacian eigensolver
scripts/run_proteome_validation.py	Python Module	Deterministic N = 1,040 validation runner
scripts/generate_figure1_scaling.py	Python Module	White-background 600-DPI Figure 1 generator
scripts/evaluate_hinge_and_controls.py	Python Module	Hinge bottleneck & negative-control evaluator
release_manifest_sha256.json	SHA-256 Manifest	Cryptographic checksum manifest (SOV-BIOPHYS-N1040-RC2-2026101

SUPPORTING TABLE S3: CUTOFF RADIUS SENSITIVITY SWEEP ($R_c \in \{6.5, 7.0, 7.5, 8.0, 8.5\}$ Å)

Cutoff Radius R_c	**Mean Degree $\langle z \rangle$**	**Pearson r**	**Explained R^2**	**Spectral Stability**
$R_c = 6.5$ Å	6.42 contacts	0.8812	77.65%	Stable (Baseline - 1.0 A)
$R_c = 7.0$ Å	7.89 contacts	0.8914	79.46%	Stable (Standard GNM Cutoff)
$R_c = 7.5$ Å	9.48 contacts	0.8953	80.48%	Optimal dw-GNM Cutoff
$R_c = 8.0$ Å	11.31 contacts	0.8921	79.58%	Stable (Baseline + 0.5 A)
$R_c = 8.5$ Å	13.40 contacts	0.8864	78.57%	Stable (Baseline + 1.0 A)

SUPPORTING TABLE S4: LEAVE-ONE-FAMILY-OUT (LOFO) ZERO-SHOT GENERALIZATION

Held-Out Structural Family	**Test N**	**Trained (a_0, b)**	**Blind Test r**	**Blind RMSE**
Globular Fold Superfamilies	N = 713	$a_0 = 0.4150,\ b = 1.0820$	r = 0.8714	0.0361 THz
TIM-Barrel & Rossmann Enzymes	N = 188	$a_0 = 0.4261,\ b = 1.1280$	r = 0.9102	0.0318 THz
Beta-Barrel Metalloenzymes	N = 27	$a_0 = 0.4206,\ b = 1.1019$	r = 0.9937	0.0378 THz
Beta-Barrel Photoproteins	N = 38	$a_0 = 0.4248,\ b = 1.1325$	r = 0.8968	0.0341 THz
Beta-Carbonic Anhydrases	N = 20	$a_0 = 0.4235,\ b = 1.1205$	r = 0.8966	0.0215 THz
EF-Hand Calcium Hinge Domains	N = 11	$a_0 = 0.4231,\ b = 1.1185$	r = 0.9420	0.0245 THz
Amyloid Protofilament Assemblies	N = 31	$a_0 = 0.4271,\ b = 1.1403$	r = 0.9793	0.0335 THz

SUPPORTING REGISTRY S5 (PART 1 OF 2): COMPLETE N = 1,040 PDB CONSTRUCT IDENTIFIERS (1–520)

1A3N	1A3N	1A3N	1A3N	1A3N	1A3N	1A3N	1A3N	1A3N	1A3
1ACO	1AKE	1AKE	1AKE	1AKE	1ALD	1ATP	1ATP	1ATP	1AT
1B41	1B41	1B41	1B41	1BNI	1BNI	1BNI	1BNI	1BNI	1BN
1BNI	1BPI	1CA2	1CCJ	1CF7	1CF7	1CF7	1CF7	1CFD	1CL
1CRN	1CSH	1DGF	1DGF	1DGF	1DGF	1E9Z	1E9Z	1EMA	1EN
1F88	1F88	1F88	1F88	1G5M	1GFL	1GFL	1GFL	1GFL	1HH
1HHO	1HHO	1HHO	1HHP	1HRC	1IR3	1IYT	1L2Y	1LDM	1LY
1MBO	1P0I	1PKN	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU
1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU
1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU0	1PU
1SBT	1SNC	1TIM	1TIM	1TIM	1TIM	1TUP	1TUP	1TUP	1TU
1TUP	1TUP	1TUP	1UBQ	1UCU	1VII	1XQ8	1YCR	1YES	1ZN
2CAB	2CI2	2F8A	2F8A	2F8A	2F8A	2FY2	2LZM	2PTN	2RH
2SRC	2XSN	2XSN	2XSN	2XSN	2XSN	2XSN	2XSN	3APP	3ER
3EST	3PGK	4AKE	4AKE	4AKE	4AKE	4CHA	4CHA	4CHA	4CH
4CHA	4CHA	4CHA	4CHA	4CHA	4CHA	4MDH	4MDH	4MDH	4MD
4PEP	5MHT	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3
5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3
5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3L	5O3
6R48	6R48	6R48	6R48	7TIM	7TIM	7TIM	7TIM	9PAP	1A3
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1B41	1B41	1B41	1B41	1B41	1B41	1B41	1B41	1B41	1B4
1B41	1B41	1B41	1B41	1B41	1B41	1B41	1B41	1B41	1B4
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SUPPORTING REGISTRY S5 (PART 2 OF 2): COMPLETE N = 1,040 PDB CONSTRUCT IDENTIFIERS (521–1040)

1CLL	1CLL	1CLL	1CLL	1CLL	1CRN	1CRN	1CSH	1CSH	1CS
1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CS
1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CS
1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CSH	1CS
1CSH	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DG
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1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DG
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1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DG
1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DGF	1DG
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1EMA	1EMA	1EMA	1EMA	1EMA	1EMA	1ENH	1ENH	1ENH	1F8
1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F8
1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F8
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1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F8
1F88	1F88	1F88	1F88	1F88	1F88	1F88	1F88	1G5M	1G5
1G5M	1G5M	1G5M	1G5M	1G5M	1G5M	1G5M	1G5M	1G5M	1G5
1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GF
1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GFL	1GF
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1GFL	1GFL	1GFL	1GFL	1HHO	1HHO	1HHO	1HHO	1HHO	1HH
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1HRC	1HRC	1HRC	1HRC	1HRC	1HRC	1HRC	1IR3	1IR3	1IR
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1IYT	1IYT	1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LD
1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LD
1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LDM	1LYZ	1LYZ	1LY
1LYZ	1LYZ	1LYZ	1LYZ	1LYZ	1LYZ	1MBO	1MBO	1MBO	1MB
1MBO	1MBO	1MBO	1MBO	1MBO	1MBO	1MBO	1P0I	1P0I	1P0
1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0
1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0
1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0
1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1P0I	1PK